

Cross-level eviction / reload test (TA slide 18 methodology)

Repeat over many independent target regions ('sets') and many trials each -- report the FRACTION of trials in each class, not a single trial. Mixed fractions -> report a behavioral bound, not a forced label (TA note on Experiment 5).

1. Calibrate
Reference single-access latency for
L2-hit, LLC-hit, DRAM classes
(this run, this machine)

2. Place target T
Repeated direct hits pull T
into L1 / L2

3. Pressure the LLC ONLY
Chase an empirically-found LLC
conflict ring (K lines, $K > \text{LLC ways}$)
that does not also touch T's L2 set

4. Reload T
Single fenced dependent access,
record latency in ticks

*Classify reload against the two boundaries
from step 1 (this is the ONE condition that matters):*

Reload = L2-hit or LLC-hit class
(latency close to calibrated
L2 / LLC medians)

-> inner copy SURVIVED
-> NON-INCLUSIVE / exclusive-like

Reload = DRAM class
(latency close to calibrated
DRAM median)

-> T was back-invalidated
-> INCLUSIVE